

Packaging Compliance AI

Comprehensive Capabilities Overview

Project portfolio summary



An AI-powered enterprise platform that automates packaging & label compliance review — from artwork upload through AI analysis, human proofing, regulatory cross-referencing, and executive reporting — with full multi-tenant access control and a supplier collaboration portal.

AI Compliance Engine

- Automated packaging/label compliance analysis with AI-flagged violations mapped to specific artwork regions
- Multi-provider AI support (OpenAI, Gemini, OpenRouter, Anthropic) with encrypted key storage and a single-active-model invariant
- Tiered AI orchestration (fast-triage vs. deep-reasoning models) with bounded automatic escalation for high-severity findings
- Background/asynchronous analysis jobs on a durable Postgres queue with retries and heartbeat-based recovery
- Legal guardrails — low-confidence AI findings are never surfaced as definitive violations (enforced in code, not prompts)
- AI usage & cost analytics via a per-call telemetry ledger with rate-card cost estimation

Document & Artwork Processing

- Direct-to-cloud artwork uploads (PDF, AI, INDD, PNG, JPG) via presigned URLs for large files
- OCR / document extraction with a provider-based architecture (OpenAI Vision default, Google Document AI selectable)
- Server-side PDF/artwork thumbnail rendering with caching
- Content-hash caching to avoid re-processing identical documents

Proofing & Review Studio

- Collaborative proofing workspace with pin/box markup, comments, @mentions, and approval decisions
- Full version control with append-only history, SHA-256 integrity hashing, and version restore
- AI-violation overlays anchored to model-provided bounding boxes
- Exportable proof PDFs with executive summary + embedded marked-up artwork, downloadable as named attachments
- Live reviewer presence and soft advisory review locks to prevent conflicting edits
- Concurrent-edit protection — reviewers can never silently overwrite each other's work

Review Assignment & Workload Engine

- Durable job queue with category-to-team routing rules
- SLA tracking, escalation stages, and monotonic escalation logic
- Specialist directory, workload balancing, and routing/escalation configuration
- Active-reviews dashboard unifying AI-review packages and specialist assignments
- Recurring review scheduling with deduplication

Regulatory Intelligence

- eCFR integration — weekly sync of curated federal regulation parts with semantic recall injected into AI analysis
- openFDA integration — recall lookups and per-category enforcement data
- Regulatory library, updates feed, and category-specific coverage
- Semantic search via pgvector — org- and supplier-scoped embeddings for compliance memory and regulation recall
- Self-hosted embedding generation (no external embedding dependency)

Language & Claims Review

- 6-layer AI copy/language review with a quality scoring system
- Dedicated Claims Compliance Engine that auto-escalates high/critical findings to reasoning-tier AI
- Approved-language glossary with change-tracking audit trail, fed back into AI review

Internal Standards & Documents

- Internal Policy & Standards engine — org policies treated as first-class compliance rules
- SOP document management with full version lineage and server-side diffing
- Policy repository, glossary, and a unified Resource Center hub
- Version publishing with row-locking and uniqueness guarantees

Supplier Collaboration

- Dedicated supplier portal with strict data isolation — suppliers can never see each other's data
- Anti-spoofing enforcement on supplier-scoped operations
- Supplier submissions, scorecards, and status lifecycle management
- Vendor scorecards and supplier detail views

Multi-Tenancy, Security & Access Control

- Role-based access control (RBAC), code-as-source-of-truth, gating both navigation and API routes
- Full multi-tenant org isolation — operations data never leaks across tenants

- Clerk authentication with org provisioning and employees-only login enforcement
- Object-level authorization (IDOR/BOLA protection) — downloads are record-authorized, not just authenticated
- API hardening: rate limiting, upload allowlists, stored-XSS mitigation, Helmet headers, SSRF-validated URLs
- Append-only audit trail with archiving

Analytics & Reporting

- Executive dashboards and reports
- Trend analysis, heatmaps, and a violations center
- Vendor scorecards and usage analytics
- Role-aware, org-scoped dashboard aggregation with resilient per-widget loading

Operations & Admin Console

- Admin console for users, teams, departments, roles, and role reassignment
- Activity monitor, audit center, and system configuration
- AI provider/model management and integrations management
- Self-lockout guards and auth-cache invalidation on role changes

Onboarding & Support

- In-app Training & Help center: guided tours, user guides, FAQ, videos, best practices
- Server-saved training progress
- Built-in support ticketing
- AI Assistant side panel — a conversational 'find the right tool' helper with a server-side tool catalog

Engineering Highlights

- Stack: React + TypeScript + Vite, Node.js + Express, PostgreSQL + Drizzle ORM + pgvector, Clerk auth, cloud object storage
- pnpm monorepo with OpenAPI-driven, type-safe client codegen (orval)
- Scalability hardening: single-flight TTL caches, clamped pagination, load-tested throughput baselines
- Notifications with per-user read/archive overlays over org-wide broadcasts
- Prod-safe vs. dev-only seed separation